

06. Pelvic Organ Prolapse Surgery and Complications

Companion reading handout for simultaneous listening.

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In the previous lecture, we built the foundation: prolapse is a symptom, function, anatomy problem. Not anatomy alone.

We move into surgery

And the first surgical lesson is the one that changes everything:

The uterus is not the support system.

The apex is.

A patient can have uterine prolapse, undergo vaginal hysterectomy, and still have the support defect that created the prolapse in the first place. The uterus may be the organ that descends, but removal of the uterus does not restore the level one support that failed.

That is why ACOG states that vaginal hysterectomy alone is not adequate treatment for uterine prolapse. When hysterectomy is done for prolapse, apical suspension should be performed to reduce recurrence.

So the dominant model for this lecture is:

Support the apex first. Then choose the least risky operation that matches the compartments, the patient's sexual goals, fertility or uterine-preservation goals, medical comorbidity, recurrence risk, mesh risk, and urinary risk.

That sentence is long because prolapse surgery is not one decision. It is a support strategy.

The indication

Surgery is indicated when prolapse is bothersome and nonsurgical treatment has failed or has been declined. That matters. Surgery is not indicated just because a stage number looks impressive. It is chosen because the patient is bothered by bulge, pressure, urinary dysfunction, bowel dysfunction, sexual dysfunction, exposure symptoms, or quality-of-life limitation, and because the expected benefit is worth the risk.

Before choosing the operation, ask the questions that actually determine the operation.

Which compartments are involved?

Is the apex involved?

How severe is the prolapse?

What urinary symptoms exist now?

What bowel symptoms exist now?

Is the patient sexually active, and does she want future vaginal intercourse?

Does she want future pregnancy?

Does she want uterine preservation?

What is her surgical risk?

What is her recurrence risk?

What is her tolerance for mesh-related risk?

And what procedures can the surgeon perform well?

That last point is not a throwaway line. Prolapse surgery is anatomy-dependent and skill-dependent. A theoretically excellent operation is not excellent if it is not the right operation for the patient or not within the surgeon's expertise.

The apex

Novak gives a useful mechanical reason for the apex-first rule. The anterior wall and apex are linked. When the anterior vaginal wall prolapses to the level of the hymen, point B equals zero, the apex is often already several centimeters inside the hymen rather than fully supported. In other words, what looks like an anterior wall problem may be partly an apical support problem.

This is why anterior repair alone can recur if the top is unsupported.

The resident reflex is:

Every prolapse operation should answer, what supports the apex?

Native-tissue vaginal apical repairs are often the backbone of primary reconstructive surgery.

The two major named procedures are uterosacral ligament suspension and sacrospinous ligament fixation.

Uterosacral ligament suspension restores high apical support by attaching the vaginal cuff, or sometimes the cervix, to the uterosacral ligaments. The anatomic advantage is

high support, close to the normal apical support axis.

The danger is the ureter.

The ureter runs close to the uterosacral ligament, especially near the posterior lateral cervix and upper vagina. If a suture kinks or incorporates the ureter, the operation may look finished while the kidney is silently obstructed.

That is the mechanism behind the cystoscopy rule. After uterosacral suspension, cystoscopy should confirm bladder integrity and ureteral efflux.

Sacrospinous fixation uses the sacrospinous ligament to support the cervix or vaginal cuff. It is usually a vaginal, extraperitoneal route. It can support a posthysterectomy vault or preserve the uterus in selected procedures.

Its tradeoffs are different. Because the attachment is posterior and lateral, it can shift the vaginal axis. It can be associated with postoperative buttock pain radiating to the thigh, often from nerve irritation or entrapment, and this often resolves spontaneously. It can also be followed by anterior compartment prolapse, partly because changing the axis and failing to address associated anterior support defects can expose the anterior wall.

ACOG summarizes the evidence clearly: uterosacral ligament suspension and sacrospinous ligament suspension for apical prolapse with native tissue are similarly effective.

In the OPTIMAL trial, two-year surgical success was about sixty four point five percent for uterosacral ligament suspension and sixty three point one percent for sacrospinous ligament fixation. Serious adverse event rates were also similar, about sixteen and a half percent in both groups.

The teaching point is not that the numbers are identical forever. Longer-term success can decline, and definitions of success vary. The teaching point is that these are both legitimate native-tissue vaginal apical repairs, and the choice depends on anatomy, risk, surgeon expertise, and patient factors.

Hysterectomy

The local Israeli position paper says this beautifully in practical terms: the uterus plays a passive role in prolapse, and removing it alone does not repair the pelvic support defect.

The rationale for hysterectomy during prolapse repair is not that the uterus itself is the support problem. It may remove the prolapsed organ, address uterine pathology, or create access to structures used for apical fixation. But if hysterectomy is performed for uterine prolapse, the cuff still needs support. That support may be McCall culdoplasty, uterosacral ligament suspension, sacrospinous fixation, or another apical support procedure.

This is a common exam and clinical trap.

If the plan is vaginal hysterectomy for prolapse, ask what supports the cuff afterward.

Anterior repair

Anterior colporrhaphy repairs anterior vaginal wall prolapse. It is used for descent of the bladder or urethral area into the vagina.

But anterior prolapse commonly coexists with apical prolapse. If the apex is not supported, the anterior repair is loaded by the same downward forces again. ACOG notes that supporting the apex during anterior repair reduces recurrent prolapse surgery.

So do not think of anterior colporrhaphy as isolated tightening of the front wall. Think of it as one part of a support plan. If the anterior wall is the visible bulge, the apex may still be the hidden driver.

Posterior repair

Posterior colporrhaphy and perineorrhaphy address posterior wall defects and perineal support. They are used for rectocele, perineal defects, and posterior compartment symptoms.

The major caution is tension.

Posterior repair should not create excessive levator ani tension, because tension can produce dyspareunia. It is not enough to make the posterior wall look tighter. The repair has to preserve comfortable vaginal function.

ACOG also notes that posterior vaginal wall prolapse repair is more effective through a transvaginal incision than through a transanal incision. The reason is that the gynecologic repair is correcting the vaginal wall support plane, not simply reducing redundant rectal mucosa.

But keep the bowel symptom trap from Lecture 5. Constipation, straining, levator spasm, anal sphincter dysfunction, and stool mechanics may coexist with a rectocele. Posterior repair may help splinting and bulge symptoms, but it should not be promised as a cure for all constipation.

Abdominal sacrocolpopexy

Sacrocolpopexy supports the vaginal apex by attaching mesh or graft from the vaginal apex to the anterior longitudinal ligament of the sacrum. It is a powerful apical support operation.

Its main clinical identity is this:

Sacrocolpopexy buys durability at the cost of mesh and abdominal or intraperitoneal risk.

ACOG describes abdominal sacrocolpopexy with synthetic mesh as having lower risk of recurrent prolapse and greater anatomic success compared with vaginal native-tissue apex repair. But it also has more complications.

Candidates may include patients with shortened vaginal length, intra-abdominal pathology, or high recurrence risk. ACOG names recurrence-risk features such as age younger than sixty, stage three or four prolapse, and BMI greater than twenty six.

This does not mean every younger patient with stage three prolapse must have sacrocolpopexy. It means the recurrence-risk conversation should be explicit.

Sacrocolpopexy complications are not theoretical. Compared with vaginal native-tissue apex repair, mesh sacrocolpopexy is associated with more ileus or small bowel obstruction, more thromboembolic phenomena, and more mesh or suture complications. In long-term CARE trial follow-up, the estimated seven-year mesh complication rate was ten point five percent.

Open sacrocolpopexy has shorter operative time in some comparisons, but minimally invasive sacrocolpopexy generally has less blood loss and shorter hospitalization. Robotic assistance may make the operation ergonomically easier and may shorten the learning curve for some surgeons, but randomized trials have not shown better short-term patient outcomes compared with conventional laparoscopy; operating time, pain, and cost were higher in some robotic trial data.

So the mature counseling is not, robotic is better, or abdominal is better, or vaginal is better.

The mature counseling is, what are we buying, what are we risking, and for this patient, is the tradeoff worth it?

Obliterative surgery

Obliterative procedures narrow, shorten, or close the vagina. The most familiar is colpocleisis. These operations are not failures of imagination. For the right patient, they are elegant, effective, and low-burden.

ACOG states that obliterative procedures are effective and should be considered first-line surgical treatment for women with significant medical comorbidities who do not desire future vaginal intercourse or vaginal preservation.

That is the key phrase:

When future vaginal intercourse is not desired.

The operation is powerful because it avoids a longer reconstructive procedure and can often be done with local or regional anesthesia. Reported objective and subjective

improvement are high, and ACOG cites low recurrent prolapse risk, low complication rates, low intensive care unit admission, and very low mortality in the relevant populations. In one older cohort, satisfaction at one year was ninety five percent.

But the decision must be explicit and respectful. The patient must understand that vaginal intercourse will no longer be possible after a complete obliterative procedure.

Le Fort colpocleisis preserves the uterus and leaves lateral drainage canals for cervical and uterine secretions. Because the uterus and cervix become difficult to access afterward, cervical cytology, HPV testing, and endometrial evaluation should be addressed preoperatively when indicated.

The local Israeli position paper makes the same practical warning: the main disadvantage, beyond loss of penetrative vaginal intercourse, is that future invasive evaluation for postmenopausal bleeding can be difficult.

Mesh

This is one of the places where residents need precise language.

The FDA order in the United States stopped the sale of transvaginal synthetic mesh products for the repair of pelvic organ prolapse.

That statement does not mean all pelvic mesh is the same.

It does not apply the same way to transvaginal mesh used for stress urinary incontinence. It does not apply the same way to transabdominal mesh used for sacrocolpopexy.

So if a patient asks, is mesh banned, the accurate answer is:

Transvaginal mesh products for prolapse repair are not currently sold with FDA approval in the United States, but this is distinct from midurethral sling mesh for stress urinary incontinence and from transabdominal mesh used in sacrocolpopexy.

The risk

Synthetic mesh and biologic grafts in prolapse surgery carry complications not seen in native-tissue repair. Mesh can be exposed in the vagina. It can contract. It can erode into the urethra, bladder, or rectum. It can cause discharge, bleeding, infection, pelvic pain, dyspareunia, and repeat operations.

ACOG reports mesh exposure around twelve percent after vaginal mesh prolapse surgery, with eight percent requiring repeat surgery for exposure up to three years after surgery. Mesh-related complications often require specialist care.

Mesh should not be routinely placed through posterior vaginal wall incisions for primary posterior wall prolapse. ACOG states that synthetic mesh or biologic grafts in

transvaginal posterior repair do not improve outcomes and add complication risk.

For anterior and apical prolapse, mesh may improve some anatomic outcomes, but the price is increased morbidity. That is why ACOG and the local Israeli paper frame vaginal mesh as selective: recurrent prolapse, high risk of recurrence, or situations where the benefit may justify the added risk, with detailed informed consent and appropriate surgical training.

Cystoscopy

Do not memorize cystoscopy as a ritual. Understand the risk it is following.

Routine intraoperative cystoscopy is recommended when the prolapse procedure carries meaningful risk of bladder or ureter injury.

This includes suspension of the vaginal apex to the uterosacral ligaments, sacrocolpopexy, anterior colporrhaphy, and mesh placement in the anterior or apical compartments.

Cystoscopy should be performed after the repair while the patient is still anesthetized. It should survey the bladder and confirm efflux from both ureteral orifices.

If there is no flow, reduced flow, or bladder injury, the problem should be addressed immediately. Delayed recognition of urinary tract injury increases morbidity.

Here is the memory line:

Cystoscopy follows the urinary tract risk.

Uterosacral suspension risks the ureter because of anatomy.

Anterior repair and anterior mesh risk the bladder because of compartment.

Sacrocolpopexy risks bladder, ureter, and intraperitoneal structures because of the dissection and fixation.

Occult stress urinary incontinence

This is another place where prolapse tricks the clinician.

Advanced anterior or apical prolapse can kink or obstruct the urethra. That kinking may hide stress urinary incontinence. After prolapse repair, the urethra is no longer kinked, and stress leakage may appear.

Therefore, all women with significant apical prolapse, anterior prolapse, or both should be evaluated preoperatively for occult stress urinary incontinence. This can be done with cough stress testing or urodynamics while the prolapse is reduced.

Reduction matters. You are trying to simulate the urinary tract after repair.

If the patient already has bothersome stress incontinence and is undergoing prolapse surgery, it is often prudent to address both problems, because no single operation adequately treats both prolapse and stress urinary incontinence.

For patients without stress incontinence symptoms, counseling is more nuanced.

A concomitant continence procedure reduces postoperative stress incontinence, but it adds risk.

In the CARE trial, women undergoing open sacrocolpopexy without preoperative stress incontinence had less postoperative stress incontinence when Burch colposuspension was added: thirty four percent with Burch versus fifty seven percent without.

In the OPUS trial, prophylactic midurethral sling at the time of vaginal prolapse surgery reduced postoperative stress incontinence: twenty four percent with sling versus forty nine percent without. But the sling group had more complications, including bladder perforation, urinary tract infection, bleeding, and transient postoperative voiding dysfunction.

This is why the counseling is staged versus concomitant.

A staged strategy avoids an unnecessary continence operation in some women, but may require a second procedure if stress leakage appears.

A concomitant strategy reduces postoperative stress leakage risk, but exposes all treated patients to the added risks of the continence procedure.

There is no honest shortcut. The patient should understand the tradeoff.

Complications more broadly

Native-tissue prolapse surgery can cause bleeding, infection, usually urinary tract infection, and transient voiding dysfunction. Less common but serious problems include rectovaginal or vesicovaginal fistula, ureteral injury, foreshortened vagina, restriction of vaginal caliber, pelvic pain, and dyspareunia.

In OPTIMAL, dyspareunia was noted in sixteen percent of women twenty four months after native-tissue prolapse surgery. This does not mean surgery should be avoided when indicated. It means sexual function belongs in the preoperative discussion, not only after a complication occurs.

Posterior repair is a good example. Excessive narrowing or levator tension can create pain with intercourse. A repair that looks anatomically successful can be a failure if it makes the patient's life worse.

Mesh complications are their own category: exposure, erosion, contracture, pain, dyspareunia, urinary or bowel involvement, and multiple procedures. Patients with

symptomatic mesh complications should be referred to clinicians with appropriate training, often female pelvic medicine and reconstructive surgery specialists.

Recurrence is also part of counseling. Recurrence can occur after any prolapse surgery. ACOG gives reported recurrence rates between six and thirty percent. Many patients with recurrent prolapse do not automatically need repeat surgery. They may monitor symptoms, use a pessary, or choose another operation depending on bother, compartment, prior surgery, and risk.

For recurrent apical prolapse after failed vaginal native-tissue suspension, options may include abdominal sacrocolpopexy, vaginal colpopexy with possible graft or mesh augmentation where appropriate, or colpocleisis when sexual goals and medical status fit. If the surgeon is not comfortable with these operations, referral is the correct management, not improvisation.

Let us apply the model

Case one.

A patient with uterine prolapse is scheduled for vaginal hysterectomy.

The resident question is not only, how will the uterus be removed? The resident question is, what will support the vaginal cuff afterward? If the answer is nothing, the plan is incomplete.

Case two.

A patient has primary symptomatic apical prolapse, average recurrence risk, and wants vaginal reconstruction without mesh.

Native-tissue vaginal apical repair is reasonable. Uterosacral and sacrospinous repairs are both evidence-supported options. The choice depends on anatomy, surgeon expertise, and risk discussion.

Case three.

A younger, sexually active patient has recurrent stage four vault prolapse and wants durable repair.

Now sacrocolpopexy enters the conversation. It may reduce recurrent bulge and improve anatomic durability, but the patient must understand mesh and abdominal risks. Durability is not free.

Case four.

A frail older patient has severe prolapse, significant cardiac disease, and no desire for future vaginal intercourse.

Colpocleisis may be the best operation, not because it is less sophisticated, but because it matches the patient's goals and medical risk. Confirm that she understands

the irreversible loss of vaginal intercourse, and address cervical and endometrial evaluation when the uterus is preserved.

Case five.

A patient with posterior prolapse asks whether surgery will cure constipation.

Be careful. If she splints because of a rectocele, repair may help. But constipation is multifactorial. Stool consistency, straining habits, pelvic floor coordination, levator spasm, and bowel disease matter. Do not promise a bowel cure from an anatomic repair.

Case six.

A patient with significant anterior and apical prolapse denies stress leakage. Reduction cough test is positive.

Now she has occult stress urinary incontinence. Counsel her about concomitant versus staged continence treatment. Explain that repair may unmask leakage, and that adding a sling or Burch-type procedure reduces that risk but adds complications.

Case seven.

A patient asks, is mesh banned?

Answer precisely. Transvaginal mesh products for prolapse repair are not currently sold with FDA approval in the United States. This is different from midurethral sling mesh for stress urinary incontinence and different from transabdominal mesh used in sacrocolpopexy. Then discuss the actual risks and alternatives for her specific operation.

Compress the lecture

Prolapse surgery is not one operation. It is a support strategy.

The first question is not, what is bulging?

The first question is, what support has failed?

The apex is the keystone. Hysterectomy alone does not repair apical support. Anterior repair without apical thinking can fail. Posterior repair must preserve sexual comfort and respect bowel physiology.

Native-tissue vaginal apical repairs are effective and avoid mesh, but recurrence is possible. Sacrocolpopexy improves anatomic durability for selected patients, but brings mesh and abdominal risks. Obliterative surgery can be an excellent first-line operation for patients with significant comorbidity who do not desire future vaginal intercourse.

Mesh counseling must be precise. Cystoscopy should follow urinary tract risk. Occult stress incontinence must be tested with the prolapse reduced before significant

anterior or apical prolapse surgery.

The final resident reflex is this:

If you operate on prolapse, answer three questions before the incision.

What supports the apex?

What happens to the bladder after the repair?

And does this operation match the patient's sexual goals, medical risk, and tolerance for recurrence or mesh complications?

That is the difference between repairing a bulge and designing a safe prolapse operation.